## Lab 4 Specifications

Lab 4 is the first introduction to the STM32L432KC microcontroller. The specifications are designed to highlight important habits of mind for working with microcontrollers, particularly understanding the relationship between the core and peripheral hardware.

### Hardware Specifications

#### Proficiency

- ☐ Design plays *Für Elise* from provided starter code.
- ☐ Note durations match the durations specified in the starter code for *Für Elise* (i.e., the tune plays at the correct tempo).
- ☐ All rests (pauses with no sound) are played properly.
- ☐ Design uses a correctly wired audio amplifier to drive the speaker.

#### Excellence

- ☐ Design contains potentiometer to control the output volume.
- ☐ Design plays an extra composition of your choice. You need not compose the tune from scratch, it is acceptable to transpose an existing tune.

### Software and Microcontroller Design Specifications

#### Proficiency

- ☐ Design makes use of at least one timer peripheral, accessed with writes to appropriate memory locations.
- ☐ Code uses `#define` macros for memory-mapped registers, copying the struct and base address design format from class.

#### Excellence

- ☐ Code does not divide by zero or commit other mathematical errors.

### Other Documentation

### Proficiency

- ☐ Documentation and calculations show how the durations and pitches are related to the timer configuration. Off by one errors are acceptable, but the calculations should be correct in principle.
- ☐ Calculations show that the system can create 220 Hz and 1000 Hz pitches within 1% accuracy. Off by one errors are acceptable, but the calculations should be correct in principle.

### Excellence

- ☐ Calculations show **minimum duration** supported
- ☐ Calculations show **maximum duration** supported
- ☐ Calculations show **minimum frequency** supported
- ☐ Calculations show **maximum frequency** supported
- ☐ Calculations show that **all notes in Für Elise** can be replicated to within 1% accuracy. Comment on whether or not checking at 220 Hz and 1000 Hz is sufficient to prove that all notes are within error bounds. Note that these calculations do not all need to be done by hand (though one example should be). Links to spreadsheets from your lab writeup are acceptable.

### Schematic Specifications

### Proficiency

- ☐ Schematic correctly represents the circuit that was built on the breadboard.
- ☐ No parts of motherboard schematic replicated. Shows only off-board components. Indicate on-board components with a brief paragraph that links the motherboard schematic (here).
- ☐ No parts internal to the chip (e.g., internal pullup resistors) drawn on schematic. Indicate important pullups, capacitors, etc. with comments instead.
- ☐ All pin names labeled.
- ☐ All pin numbers labeled.
- ☐ Crossing wires clearly identified as junction or unconnected.
- ☐ Neat layout (e.g., clear organization and spacing).
- ☐ All parts labeled with part number.
- ☐ All component values present.

### Excellence

- ☐ Standard symbols used for all components where applicable.
- ☐ Signals "flow" from left to right where possible (e.g., inputs on left hand side, outputs on right hand side, power on top, ground on bottom)
- ☐ Title block with author name, title, and date.

### Lab Writeup

### Proficiency and Excellence

- ☐ Documentation embedded in a well-structured, concise report with one section per major deliverable. Sections should match structure of other specs.
- ☐ Spoiler tags are used to hide screenshots and code. Some unspoilered entries (~1 waveform, essential code snippets) allowed if they are essential to the narrative, kept to a minimum.
- ☐ Writeup contains minimal spelling or grammar issues and any errors do not significantly detract from clarity of the writeup.
- ☐ Statement of whether the design meets all the requirements. If not, list the shortcomings.
- ☐ AI prototype attempted and some reflection is recorded.
- ☐ Number of hours spent working on the lab are included.
- ☐ (Optional) List comments or suggestions on what was particularly good about the assignment or what you think needs to change in future versions.